Protocol H9X-MC-GBGL

A Randomized, Double-Blind, Parallel Arm Study of the Efficacy and Safety of Investigational Dulaglutide Doses When Added to Metformin in Patients with Type 2 Diabetes Mellitus

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(AWARD-11: Assessment of Weekly AdministRation of LY2189265 in Diabetes-11)

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Dulaglutide (LY2189265)

Study H9X-MC-GBGL is a Phase 3, randomized, double-blind trial designed to assess the efficacy and safety of once weekly investigational dulaglutide doses (4.5 mg or 3.0 mg) compared to dulaglutide 1.5 mg in patients with type 2 diabetes mellitus on metformin monotherapy.

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on approval date provided below.

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1. Synopsis

Title of Study: A Randomized, Double-Blind, Parallel-Arm Study of the Efficacy and Safety of Investigational Dulaglutide Doses When Added to Metformin in Patients with Type 2 Diabetes Mellitus

Rationale:

Study H9X-MC-GBGL (GBGL) is a Phase 3 trial designed to assess the safety and efficacy of once weekly dulaglutide 3.0 mg and 4.5 mg in comparison to dulaglutide 1.5 mg (the highest dose approved by regulatory agencies). The overall goal of the Phase 3 program is to gain regulatory approval for inclusion of at least 1 higher dose in the Trulicity label for treatment of patients with type 2 diabetes mellitus (T2D) who require additional glucose-lowering to obtain their glycemic target. The additional dosing option(s) will be administered using the same device as that used for currently approved doses of 0.75 mg and 1.5 mg (single-dose pen [SDP]).

Objectives/Endpoints:

Objectives	Endpoints
Primary To demonstrate that once weekly dulaglutide 4.5 mg and/or 3.0 mg is superior to dulaglutide 1.5 mg for change from baseline in HbA1c at 36 weeks in patients with inadequately controlled T2D on concomitant metformin therapy.	• The change in HbA1c from baseline
Secondary *Efficacy:* To demonstrate that once weekly dulaglutide 4.5 mg and/or 3.0 mg is superior to dulaglutide 1.5 mg for secondary efficacy parameters at 36 weeks (controlled for Type 1 error).	• The change in body weight from baseline • Proportion of patients achieving HbA1c target <7.0% (53 mmol/mol) • The change in fasting serum glucose (FSG) from baseline
Safety: To compare each investigational dulaglutide arm (4.5 mg, 3.0 mg) to the 1.5 mg arm for selected safety parameters through 36 and 52 weeks (unless noted otherwise)	• Incidence of treatment-emergent adverse events (TEAEs) and discontinuation of study drug due to adverse events (AEs) • Adjudicated and confirmed cardiovascular and pancreatic AEs • Incidence of thyroid neoplasm AEs • Incidence of treatment-emergent (TE) dulaglutide anti-drug antibodies (ADA) and systemic hypersensitivity reactions • Change from baseline in pulse rate (PR) • Electrocardiogram (ECG) parameters • Occurrence of hypoglycemic episodes

Objectives	Endpoints
Pharmacokinetics (PK) and Pharmacodynamics (PD) To characterize dulaglutide PK and the dose and/or exposure-response relationships for key efficacy (eg, HbA1c and weight), and safety (eg, heart rate) endpoints.	• PK parameters (eg, maximum concentration [Cmax], area under the curve [AUC]) at steady state • Pharmacodynamic evaluations may include changes from baseline in HbA1c, body weight, and heart rate at Weeks 36 and 52

Abbreviations: HbA1c = hemoglobin A1c; T2D = type 2 diabetes mellitus.

Summary of Study Design:

Study GBGL is a Phase 3, multicenter, randomized, double-blind, parallel-arm study with 3 study periods (Lead-In, Treatment, and Safety Follow-Up) in patients with T2D with inadequate glycemic control on metformin only; the study will have 52-week treatment duration, with primary endpoint at 36 weeks.

Treatment Arms and Duration:

At Visit 3 patients will be randomized in a 1:1:1 ratio to weekly injections of dulaglutide 4.5 mg, 3.0 mg, or 1.5 mg, in combination with stable doses of metformin. Patients will be titrated through sequential 4-week treatment segments beginning with 0.75 mg once weekly followed by 1.5 mg once weekly. At Week 8, patients randomized to the dulaglutide 1.5 mg group will continue on this dose for the remainder of the Treatment Period. Patients randomized to the dulaglutide 3.0 mg group will be escalated to 3.0 mg once weekly at Week 8 and will continue on this dose for the remainder of the Treatment Period. Patients assigned to the dulaglutide 4.5 mg group will be escalated to 3.0 mg once weekly at Week 8 for 4 weeks, followed by escalation to their final dose of 4.5 mg once weekly at Week 12. Study participants will be treated for 52 weeks, with the primary objectives assessed at Week 36. A 4-week Safety Follow-Up Period will follow the completion of the Treatment Period.

Number of Patients:

A sample size of approximately 1800 patients assuming 15% dropout rate will be enrolled (randomized) in order to obtain approximately 510 completers per arm at 36 weeks. This sample size provides at least 80% power for demonstrating superiority of at least 1 dulaglutide higher dose to dulaglutide 1.5 mg in glycated hemoglobin (HbA1c) change from baseline at Week 36. This assumes a true treatment difference in HbA1c of -0.22% with a standard deviation (SD) of 1.1%.

Statistical Analysis:

Efficacy Analyses:

There will be 2 primary estimands to compare the investigational dulaglutide arms (4.5 mg and 3.0 mg) with dulaglutide 1.5 mg in terms of the primary measure of HbA1c change from baseline to 36 weeks. One primary estimand will be an efficacy estimand (*de jure* effect) which will use the data collected before initiation of any rescue medication or premature treatment discontinuation to demonstrate the effect of treatment and avoid confounding effects of other antihyperglycemic agents; the other primary estimand will be a treatment-regimen estimand (*de facto* effect) which will include data collected after initiation of other antihyperglycemic therapy and/or after premature treatment discontinuation. The efficacy estimand measures the benefit of treatment when taken as

directed, and the treatment-regimen estimand measures the benefit of treatment as actually taken (that is, irrespective of adherence to investigational product or introduction of other antihyperglycemic therapy). Both estimands will use the same primary analysis model and each will be tested at the full significance level of 0.05. The treatment-regimen estimand is included as primary at the request of the US Food and Drug Administration (FDA). The efficacy estimand will be considered primary for all other purposes. The primary analysis model for HbA1c will be a mixed-model for repeated measures (MMRM) using restricted maximum likelihood, with pooled country, treatment, visit, and treatment-by-visit as fixed effects, and baseline HbA1c as a covariate. An unstructured covariance structure will be used to model the within-patient errors. The primary analysis model will be repeated using all available data from the Per Protocol and Completers populations to check the robustness of the analyses.

Multiple imputation for missing data within the same treatment arm will be conducted based on data from retrieved dropouts. Retrieved dropouts are defined as patients who had their HbA1c value at the Week 36 visit assessed after early discontinuation of study drug and/or initiation of rescue medication. An analysis of covariance (ANCOVA) model for the treatment-regimen estimand will be fit to the complete datasets, containing the same covariates as the primary analysis, by removing the visit-related terms.

Analyses of other continuous secondary efficacy measures will be performed using MMRM in the Intent-to-Treat (ITT) Population. For percentages of patients achieving target HbA1c <7.0%, longitudinal logistic regression with repeated measures will be conducted.

The primary and key secondary analyses will follow a graphical approach for multiple comparisons to strongly control the overall Type 1 error rate in the trial at a 2-sided alpha of 0.05.

Safety Analyses:

Summary statistics will be provided for incidence of treatment-emergent adverse events (TEAEs), serious adverse events (SAEs), and study discontinuation due to adverse events (AEs) or death during the Treatment Period. Counts and proportions of patients experiencing AEs will be reported for each treatment group, and Fisher's exact test will be used to compare the treatment groups.

For continuous laboratory analytes, the change from baseline to endpoint will be analyzed using an analysis of variance (ANOVA) on the rank-transformed data, with treatment as a fixed effect. Last observation carried forward (LOCF) will be used to impute missing postbaseline values. For qualitative laboratory analytes, counts and percentages of patients with normal and abnormal values will be analyzed using Fisher's exact test.

Treatment differences in incidence of hypoglycemic episodes will be assessed by Fisher's exact test. Treatment differences in rates of hypoglycemic episodes may be assessed by a likelihood-based approach for repeated measures with negative binomial distribution if data warrant; otherwise, the Wilcoxon rank-sum test will be used.

2. Schedule of Activities

	Screening and Lead-In		Treatment Period										Safety Follow-Up
Visit	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**ET[a]**	**801**
Week of Treatment	**-2**	**-1**	**1**	**4**	**8**	**12**	**18**	**26**	**36**	**44**	**52**		**56**
Allowable Deviation (Days)[b]	**±3**	**±3**		**±3**	**±3**	**±3**	**±7**	**±7**	**±7**	**±7**	**±7**		**±7**
Informed consent	**X**												
Randomization[c]			**X**										
Clinical Assessments													
Medical History (General)	**X**												
CV risk factor history		**X**											
Gallbladder disease history		**X**											
Physical	**X**												
Height	**X**												
Weight	**X**		**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Vital Signs (3 sitting BP and PR measurements)[d]	**X**		**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Electrocardiogram	**X**					**X**[e]	**X**		**X**[e]		**X**[e]	**X**[e]	**X**[e]
Adverse events		**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Concomitant medications	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Review/record hypoglycemic events			**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Remind patients about 6-point SMPG		**X**						**X**		**X**			
Record 6-point SMPG values[f]			**X**						**X**		**X**	**X**[g]	
Patient summary											**X**	**X**	**X**
Patient Education and Management													
Dispense study diary, instruct in use		**X**	**X**			**X**	**X**		**X**	**X**	**X**		
Review study diary			**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Return study diary			**X**			**X**	**X**		**X**	**X**	**X**	**X**	**X**
Diabetes management education[h]		**X**											
Dispense PG meter/supplies		**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**		
PG meter, SMPG training[h]		**X**	**X**										
Review SMPG values			**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Injection training[h]		**X**	**X**										
Dispense study drugs and/or injection supplies			**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**			

Schedule of Activities

	Screening and Lead-In		Treatment Period										Safety Follow-Up
Visit	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**ET[a]**	**801**
Week of Treatment	**-2**	**-1**	**1**	**4**	**8**	**12**	**18**	**26**	**36**	**44**	**52**	**ET**	**56**
Allowable Deviation (Days)[b]	**±3**	**±3**		**±3**	**±3**	**±3**	**±7**	**±7**	**±7**	**±7**	**±7**		**±7**
Patient Education and Management (concluded)													
Observe patient inject study drug[i]			**X**										
Patient returns study drugs and/or injection supplies				**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	
Assess study drug compliance				**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	
Laboratory Tests													
Serum pregnancy test[k]	**X**												
Urine pregnancy test[l]			**X**			**X**		**X**			**X**		
Follicle-stimulating hormone test[m]	**X**												
Chemistry panel	**X**								**X**		**X**	**X**	**X**
Lipid panel (fasting)[n]			**X**						**X**		**X**	**X**	
Urinalysis	**X**								**X**		**X**	**X**	
Albumin/creatinine ratio (urine)[o]	**X**								**X**		**X**	**X**	
eGFR (CKD-EPI)[p]	**X**								**X**		**X**	**X**	**X**
Cystatin C			**X**						**X**		**X**	**X**	
Hematology	**X**								**X**		**X**	**X**	
Immunogenicity			**X**	**X**		**X**			**X**		**X**	**X**	**X**
Pancreatic amylase, lipase	**X**		**X**						**X**		**X**	**X**	**X**
Calcitonin	**X**					**X**			**X**		**X**	**X**	
HbA1c	**X**		**X**		**X**	**X**		**X**	**X**		**X**	**X**	**X**
Glucose (fasting)[n]			**X**		**X**	**X**		**X**	**X**		**X**	**X**	**X**
Serum insulin (fasting)[n]			**X**						**X**		**X**	**X**	
C-peptide			**X**						**X**		**X**	**X**	
Plasma glucagon (fasting)[n,q]			**X**						**X**		**X**	**X**	
Pharmacogenetic (stored sample)			**X**[r]										
Nonpharmacogenetic biomarker (plasma and serum) (stored sample)			**X**						**X**		**X**	**X**	
PK Samples													
PK samples[s]				**X**		**X**			**X**		**X**	**X**	**X**

Schedule of Activities

	Screening and Lead-In		Treatment Period										Safety Follow-Up
Visit	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**ET[a]**	**801**
Week of Treatment	**-2**	**-1**	**1**	**4**	**8**	**12**	**18**	**26**	**36**	**44**	**52**	**ET**	**56**
Allowable Deviation (Days)[b]	**±3**	**±3**		**±3**	**±3**	**±3**	**±7**	**±7**	**±7**	**±7**	**±7**		**±7**
PRO Questionnaires (Health Economics)													
Diabetes Injection Device Experience Questionnaire (DID-EQ)						**X**						**X[t]**	
EQ-5D-5L			**X**						**X**		**X**	**X**	
Impact of Weight on Self-Perceptions Questionnaire (IW-SP)			**X**						**X**		**X**	**X**	
Ability to Perform Physical Activities of Daily Living Questionnaire (APPADL)			**X**						**X**		**X**	**X**	

Abbreviations: BP = blood pressure; CKD-EPI = Chronic Kidney Disease-Epidemiology Collaboration equation; CV = cardiovascular; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; ET = early termination; HbA1c = hemoglobin A1c; OAM = oral antihyperglycemic medication; PG = plasma glucose; PK = pharmacokinetic; PR = pulse rate; PRO = Patient-Reported Outcomes; SMPG = self-monitored plasma glucose.

a Patients who are unable or unwilling to continue in the study for any reason will perform an ET visit. If the patient is discontinuing during an unscheduled visit, that visit should be performed as the ET visit. If the patient is discontinuing during a scheduled visit, that visit should be performed as an ET visit. Visit 801 (Safety Follow-Up Visit) should be performed 4 weeks after the ET visit as the final study visit.

b The visit date is determined in relation to the date of the randomization visit (± the allowed visit window).

c Baseline assessments must be completed before processing in the interactive web-response system (IWRS).

d Vital sign measurements should be taken before obtaining an ECG tracing and before collection of blood samples for laboratory testing, at visits where required. The participant should sit quietly for 5 minutes before vital sign measurements are taken. For each parameter, 3 measurements will be taken using the same arm; the recordings should be taken at least 1 minute apart.

e Electrocardiograms occurring on visits with PK collection should be collected at least 30 minutes prior to obtaining the sample for PK measurement.

f 6-point SMPG consists of measurements before and 2 hours after each of 3 main meals within the same day. These SMPG profiles will be collected by the patient within 1 week prior to the assigned visits. If 6-point SMPG is not performed, then data from 4-point SMPG can be used.

g If performed prior to visit; if 6-point SMPG is not performed, then data from 4-point SMPG can be used.

h All training should be repeated as needed to ensure patient compliance.

i Patients should administer their first dose of study drug at the end of this visit, after other study procedures and randomization.

j Study drug compliance will be reviewed at every visit between Visit 4 and Visit 11.

k A serum pregnancy test will be performed at Visit 1 for women of childbearing potential only.

Schedule of Activities

l A urine pregnancy test will be performed at Visit 3, with the result available prior to randomization and first injection of study drug, and at additional visits as shown in the schedule, for women of childbearing potential only. Additional pregnancy tests may be performed at the investigator's discretion or if required per local regulations and/or institutional guidelines during the study.

m Follicle-stimulating hormone test will be collected at Visit 1 for postmenopausal women at least 50 years of age with an intact uterus, not on hormone therapy, and who have had spontaneous amenorrhea for more than 6 months and less than 12 months (unless specified otherwise by local regulatory requirements). After Visit 1, additional tests may be performed at the investigator's discretion during the study.

n For Visit 3, Visit 5, Visit 6, Visit 8, Visit 9, Visit 11, Visit 801, and the ET visit, patients should be reminded to report to the site in a fasting condition, after a period of approximately 8 hours without eating, drinking (except water), or any significant physical activity and before taking their OAMs (if used).

o Urinary albumin and creatinine will be measured and the ratio will be calculated.

p The CKD-EPI equation will be used by the central laboratory to compute and report eGFR. Values that are <30 mL/min/1.73 m^2 at Visit 1 will be required to be retested at Visit 2. If the highest of the 2 values is lower than <30 mL/min/1.73 m^2 (or lower than the cutoff value for discontinuation of metformin per country-specific label), the patient will be discontinued from the trial prior to randomization. After randomization (Visit 3), values that would require clinical action per protocol or country-specific label must be first confirmed on a retest before any treatment adjustment is implemented.

q Samples should be collected in P800 tube. Aliquot samples, 1 for glucagon and the remainder for nonpharmacogenetic biomarker.

r The pharmacogenetic sample should be collected at Visit 3, or at the earliest visit after obtaining local approval.

s The PK sample should be collected within 30 min after the ECG is obtained on the visits where this procedure is also performed (Visits 6, 9, 11, ET, and 801). At Visit 4, the PK sample can be collected at any time during the clinic visit. The date and time of administration of all dulaglutide doses, and in particular the last dulaglutide dose administered prior to each PK sample, together with the date and time of the PK sample, must be clearly recorded.

t DID-EQ will be administered at the ET visit only if ET occurs at or before Week 12 (before Visit 6).

3. Introduction

3.1. Study Rationale

Dulaglutide (Trulicity®) is a once weekly glucagon-like peptide-1 (GLP-1) receptor agonist (GLP-1 RA) that is approved for use as an adjunct to diet and exercise to improve glycemic control in patients with type 2 diabetes (T2D). Treatment of patients with T2D with dulaglutide results in significant reductions in glycated hemoglobin (HbA1c) with low risk of hypoglycemia, and reductions in body weight. Two once weekly dulaglutide doses, 0.75 mg and 1.5 mg, were studied in the Phase 3 development program and received regulatory approval in the United States (US) and the European Union in 2014 (Trulicity US Package Insert [USPI], Summary of Product Characteristics [SmPC]).

While therapy with currently approved doses of dulaglutide enabled the majority of patients included in the Phase 3 program to attain their glycemic targets (sometimes with use of other concomitant medications for T2D), 30%-50% of patients receiving approved therapies today, including dulaglutide, are not reaching glycemic control goals (Stark Casagrande et al. 2013; Jendle et al. 2016). Therefore, there remains an important medical need to provide enhanced efficacy of pharmaceutical agents while also preserving an overall acceptable benefit/risk profile.

3.2. Background

Pharmacokinetic (PK)/pharmacodynamics (PD) modeling from original Phase 1-2 dulaglutide studies suggested a dose-dependent incremental benefit with dulaglutide versus currently approved doses on HbA1c and body weight reduction. Phase 2 Study H9X-MC-GBGJ (GBGJ) was recently conducted to assess the safety and efficacy of once weekly dulaglutide 3.0 mg and 4.5 mg in comparison to placebo in 317 patients with T2D treated with metformin monotherapy, with exploratory comparisons to dulaglutide 1.5 mg (the highest dose approved by regulatory agencies). Data from Study GBGJ provided evidence for incremental efficacy of these higher doses of dulaglutide versus 1.5 mg once weekly on glycemic control and body weight reduction, particularly at higher baseline HbA1c levels, with an acceptable tolerability and safety profile at both higher doses. The trial results also suggested stepwise titration of dulaglutide dose was associated with acceptable tolerability of the 2 higher doses.

Further study of higher doses of dulaglutide also was assessed in context with prior dose-finding studies. The currently approved dulaglutide doses (0.75 mg and 1.5 mg once weekly) were selected in the Phase 2/3 Study H9X-MC-GBCF (GBCF), which initially included doses up to 3.0 mg once weekly (Skrivanek et al. 2014). The purpose of the first stage of Study GBCF was to identify an optimal dose using prespecified measures of efficacy (HbA1c and weight) and safety (diastolic blood pressure [DBP] and heart rate [HR]). Following a number of interim data reviews, the Data Monitoring Committee (DMC) recommended discontinuing the dulaglutide 3.0 mg dose because of higher incidence in pancreatic enzyme values above upper limit of normal (ULN) (with or without abdominal pain or suspected pancreatitis), increases in HR, and higher incidence of gastrointestinal (GI) adverse events (AEs) (Skrivanek et al. 2014).

Since original dose selection and subsequent approval of dulaglutide, additional data have provided a better understanding of the clinical significance of the findings which prompted the DMC to terminate the 3.0 mg dose in Study GBCF. Elevations in pancreatic enzymes associated with GLP-1 RA treatment have not been associated with adverse clinical outcomes based on the results of adjudication (Nauck et al. 2017; Steinberg et al. 2017a; Steinberg et al. 2017b), and the risk of adjudicated pancreatitis and investigator-reported pancreatitis in patients treated with dulaglutide was similar to the risk observed with nonincretin comparators in dulaglutide registration trials (Nauck et al. 2017). Heart rate elevations are now recognized to be a class effect common to all GLP-1 RAs (Lorenz et al. 2017). Despite increases in HR, completed cardiovascular (CV) outcome trials for GLP-1 RAs have shown either neutral to favorable effects on the risk of major CV events compared to placebo in patients with T2D and high CV risk (Pfeffer et al. 2015; Marso et al. 2016a; Marso et al. 2016b; Holman et al. 2017). Finally, it is now understood that a slow, stepwise dose titration of weekly GLP-1 RAs upon initiation can attenuate the occurrence and severity of GIAEs at higher doses, providing for greater glucose-lowering effects and more robust body weight reduction (Nauck et al. 2016).

Thus, based on the limited amount of efficacy data available for higher dulaglutide doses, the promising results of Phase 2 Study GBGJ, and considering the updated understanding of the benefit/risk profile of GLP-1 RAs as a class, a re-evaluation of the optimal weekly dose of dulaglutide for glycemic control was deemed warranted. The purpose of this Phase 3 study (H9X-MC-GBGL [GBGL]) is to assess the safety and efficacy of once weekly dulaglutide 3.0 mg and 4.5 mg in comparison to dulaglutide 1.5 mg. The overall goal of the Phase 3 program is to gain regulatory approval for inclusion of at least 1 higher doses in the Trulicity label for treatment of patients with T2D who require additional glucose lowering to obtain glycemic targets, to be administered using the same device as that used for currently approved doses of 0.75 mg and 1.5 mg (single-dose pen [SDP]).

3.3. Benefit/Risk Assessment

More information about the known and expected benefits, risks, serious adverse events (SAEs) and reasonably anticipated AEs of dulaglutide may be found in the Investigator's Brochure (IB).

In addition, detailed information about the known and expected benefits and risks of dulaglutide may be found in the Trulicity USPI (2017 [WWW]) and SmPC (2017 [WWW]).

4. Objectives and Endpoints

Table GBGL.1 presents the objectives and endpoints of the study.

Table GBGL.1. Objectives and Endpoints

Objectives	Endpoints
Primary To demonstrate that once weekly dulaglutide 4.5 mg and/or 3.0 mg is superior to dulaglutide 1.5 mg for change from baseline in HbA1c at 36 weeks in patients with inadequately controlled T2D on concomitant metformin therapy.	• The change in HbA1c from baseline
Secondary *Efficacy:* To demonstrate that once weekly dulaglutide 4.5 mg and/or 3.0 mg is superior to dulaglutide 1.5 mg for secondary efficacy parameters at 36 weeks (controlled for Type 1 error). *Safety:* To compare each investigational dulaglutide arm (4.5 mg, 3.0 mg) to the 1.5 mg arm for selected safety parameters through 36 and 52 weeks (unless noted otherwise) *Pharmacokinetics (PK) and Pharmacodynamics (PD):* To characterize dulaglutide PK and the dose and/or exposure-response relationships for key efficacy (eg, HbA1c and weight), and safety (eg, heart rate [HR]) endpoints.	• The change in body weight from baseline • Proportion of patients achieving HbA1c target <7.0% (53 mmol/mol) • The change in fasting serum glucose (FSG) from baseline • Incidence of treatment-emergent adverse events (TEAEs) and discontinuation of study drug due to adverse events (AEs) • Adjudicated and confirmed cardiovascular and pancreatic AEs • Incidence of thyroid neoplasm AEs • Incidence of treatment-emergent (TE) dulaglutide anti-drug antibodies (ADA) and systemic hypersensitivity reactions • Change from baseline in pulse rate (PR) • Electrocardiogram (ECG) parameters • Occurrence of hypoglycemic episodes • PK parameters (eg, maximum concentration [Cmax], area under the curve [AUC]) at steady state • Pharmacodynamic evaluations may include changes from baseline in HbA1c, body weight, and heart rate at Weeks 36 and 52
Tertiary/Exploratory To compare once weekly dulaglutide 4.5 mg and 3.0 mg to the 1.5 mg arm as measured by the primary and secondary outcome measures at 52 weeks.	• The change in HbA1c from baseline • The change in body weight from baseline • Proportion of patients achieving HbA1c target

Objectives	Endpoints
To compare once weekly dulaglutide 4.5 mg and 3.0 mg to the 1.5 mg arm for exploratory measures through 36 and 52 weeks (unless noted otherwise).	<7.0% (53 mmol/mol) • The change in FSG from baseline • Proportion of patients achieving HbA1c target ≤6.5% (48 mmol/mol) • Change from baseline in 6-point self-monitored plasma glucose (SMPG) profile • Proportion of patients achieving ≥5% body weight loss • Proportion of patients achieving ≥10% body weight loss • Proportion of patients meeting the composite endpoint of HbA1c <7.0% (53 mmol/mol), no weight gain, and no documented symptomatic or severe hypoglycemia • Proportion of patients meeting the composite endpoint of HbA1c <7.0% (53 mmol/mol), body weight loss ≥5%, and no documented symptomatic or severe hypoglycemia • Changes from baseline in fasting plasma glucagon, HOMA2-%B, HOMA2-IR, and C-peptide • Changes from baseline in serum cystatin C and cystatin C-based assessment of eGFR • Incidence of initiation of rescue therapy for severe, persistent hyperglycemia • Changes from baseline in systolic blood pressure (SBP), diastolic blood pressure (DBP), and rate pressure product (RPP) • Changes from baseline in serum lipid parameters (total cholesterol, high density lipoproteins [HDL], low density, lipoproteins [LDL], very low density lipoproteins [VLDL], and triglycerides) • Diabetes Injection Device Experience Questionnaire (DID-EQ) scores at Week 12 • Changes from baseline in EQ-5D-5L scores • Changes from baseline in Impact of Weight on Self-Perception Questionnaire (IW-SP) scores • Changes from baseline in Ability to Perform Physical Activities of Daily Living (APPADL) scores

Abbreviations: eGFR = estimated glomerular filtration rate; HbA1c = glycated hemoglobin A1c; HOMA2-%B = β-cell function as measured by the Homeostasis Model Assessment-2 method; HOMA2-IR = insulin resistance as measured by the HOMA2 method; T2D = type 2 diabetes.

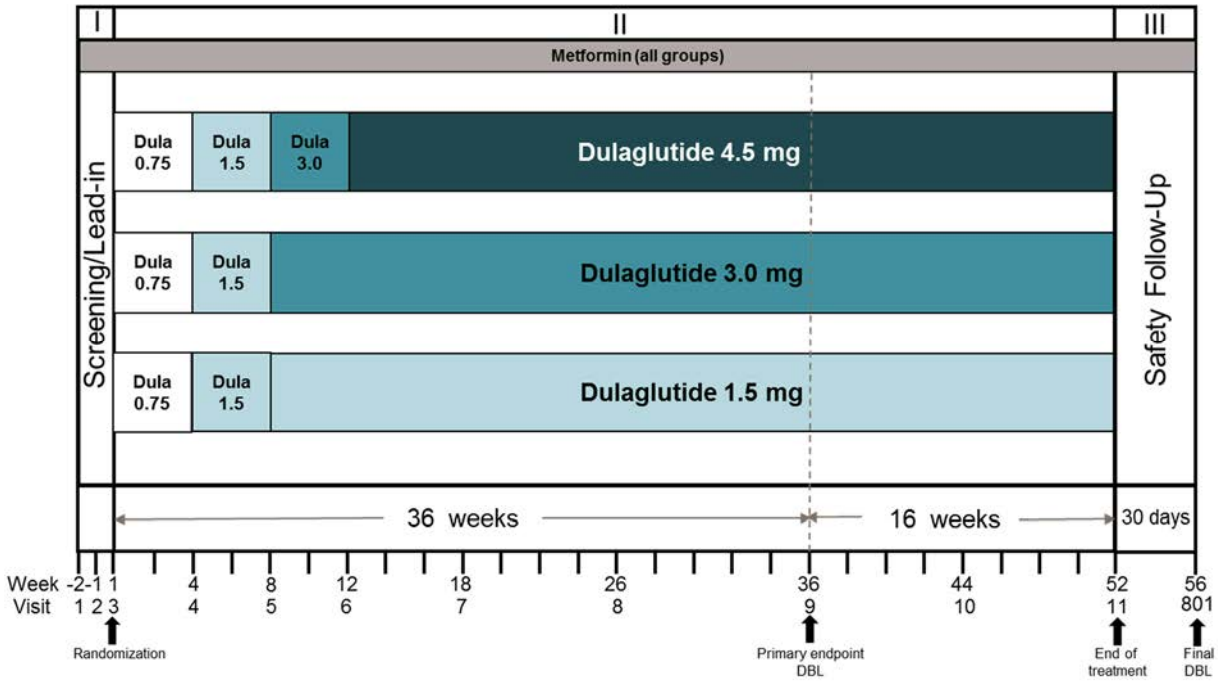
5. Study Design

5.1. Overall Design

Study H9X-MC-GBGL (GBGL) is a 52-week, Phase 3, double-blind, multicenter, parallel-arm study. The study is designed to assess the efficacy and safety of once weekly dulaglutide 3.0 mg and 4.5 mg in comparison to once weekly dulaglutide 1.5 mg following a primary 36-week treatment period and at the end of the 52-week Treatment Period. The primary objective of this trial is to demonstrate that once weekly dulaglutide 4.5 mg and/or 3.0 mg is superior to dulaglutide 1.5 mg as measured by change from baseline in HbA1c at Week 36 in patients with T2D who are inadequately controlled on concomitant metformin therapy (see Section 4). Secondary objectives (controlled for Type 1 error) are to demonstrate superiority of at least 1 of the higher dulaglutide doses to dulaglutide 1.5 mg at Week 36 on changes in body weight, proportions of patients achieving HbA1c <7.0% (53 mmol/mol), and changes in fasting serum glucose (FSG) from baseline at Week 36.

The study will consist of 3 periods: an approximately 2-week Lead-In Period followed by a 52-week Treatment Period (primary efficacy endpoint at Week 36) and a 4-week Safety Follow-Up Period.

Figure GBGL.1 illustrates the study design.



Abbreviations: DBL = database lock; dula = once weekly dulaglutide.
Note: all dulaglutide doses are once weekly.

Figure GBGL.1. Illustration of study design for Clinical Protocol H9X-MC-GBGL.

Study Period I (Screening and Lead-In):

Screening (Visit 1)

The purpose of screening procedures at Visit 1 is to establish initial eligibility and to obtain blood samples for laboratory assessments needed to confirm eligibility at Visit 2. Patients who meet all applicable inclusion criteria and none of the applicable exclusion criteria at Visit 1 will continue on their prestudy therapy until Visit 2.

Lead-In (Visit 2 to Visit 3)

At Visit 2, the screening laboratory results will be reviewed and definite patient eligibility will be established. During the Lead-In Period, eligible patients should continue their prestudy therapy (metformin, same formulation and dose), in order to allow reliable assessment of HbA1c at baseline (Visit 3). If patients develop a condition that is a contraindication for the use of metformin or initiate other agents that are prohibited between entry (Visit 1) and randomization, they will be considered ineligible and will be discontinued from the trial before randomization. During this period, patients will be trained on disease monitoring and disease management procedures, study diaries, and study procedures.

Study Period II (Treatment Period):

Randomization (Visit 3)

At Visit 3, patients will perform all required baseline study procedures (including the collection of all baseline laboratory measures) prior to randomization and prior to taking the first dose of study drug. Patients who continue to be eligible will then be randomized in a 1:1:1 ratio to 1 of the treatment arms: (1) dulaglutide 4.5 mg, (2) dulaglutide 3.0 mg, or (3) dulaglutide 1.5 mg. Immediately after randomization, the patient will inject the first dose of study drug at the study site, according to the dose escalation regimen described below. The date and time of the first dose of study drug should be recorded on the electronic case report form (eCRF).

Following randomization, patients will participate in a 52-week Treatment Period.

Postrandomization Treatment Period (end of Visit 3 to Visit 11)

Beginning at randomization, all patients will receive dulaglutide 0.75 mg once weekly for 4 weeks (Weeks 1-4), followed by dulaglutide 1.5 mg once weekly for 4 weeks (Weeks 5-8). At Week 8 the dosing will diverge based on the randomized treatment arm:

- Dulaglutide 1.5 mg arm: Patients will continue to receive dulaglutide 1.5 mg once weekly for the remainder of the treatment period.
- Dulaglutide 3.0 mg arm: Patients will receive dulaglutide 3.0 mg once weekly for the remainder of the treatment period.
- Dulaglutide 4.5 mg arm: Patients will receive dulaglutide 3.0 mg once weekly for 4 weeks (Weeks 9-12), and then escalate to dulaglutide 4.5 mg once weekly for the remainder of the treatment period.

Study Period III (Safety Follow-Up Period):

Safety Follow-Up (Visit 801) Visits: All patients who complete the treatment period are required to complete Visit 801, a safety follow-up visit approximately 4 weeks after their last visit. Patients discontinuing the study early and performing an early termination (ET) visit will also be asked to perform the safety follow-up visit, so that the safety follow-up visit will be their final visit. During the Safety Follow-Up Period, patients will not receive study drug. Patients will be treated with another glucose-lowering intervention decided upon by the investigator. Glucagon-like peptide-1 RAs (eg, exenatide, liraglutide, or commercial Trulicity) and dipeptidyl peptidase-4 (DPP-4) inhibitors are not allowed during this period. Initiation of new antihyperglycemic therapy for the Safety Follow-Up Period will not be classified as "rescue therapy" (Section 7.7.1.2). All antihyperglycemic medications will be entered on the eCRF specified for this purpose. Patients are also required to return any remaining study diaries to the study site at the end of this period.

Study Procedures

Patients will perform study procedures listed in the Schedule of Activities (Section 2).

Patients will continue to use concomitant metformin throughout the treatment period: discontinuation or changes to dose are not permitted, except in the situations where dose adjustment or complete discontinuation is required per country-specific label or when allowed per study protocol (for further details see Section 7.7.1.1).

Patients will be instructed to perform fasting plasma glucose (FPG) measurements once daily and 4-point self-monitored plasma glucose (SMPG) measurements (consisting of fasting, prelunch, predinner, and bedtime plasma glucose [PG] measurements) once weekly and to record all results in diaries; these results will be used for glucose management only. In addition, 6-point SMPG (prior to and 2 hours after the morning, midday, and evening meals) will be collected during the week preceding prespecified clinic visits as shown in the Schedule of Activities (Section 2) and will be used for efficacy analyses. During the weeks when the 6-point SMPG profiles are to be collected, patients will not be required to collect 4-point SMPG profiles, and the 6-point profiles will be used for glucose management instead. Data from the 6-point profiles will be entered into the eCRF as shown on the Schedule of Activities (Section 2). In the event that a 6-point SMPG profile is not collected as required, then data from the 4-point SMPG collected during the week preceding the required visit may be entered in the eCRF instead.

Patients who develop severe, persistent hyperglycemia based on prespecified thresholds (see Section 7.7.5.3) will receive a new glucose-lowering intervention ("rescue therapy") as outlined Section 7.7.1.2 and will also continue to administer study drug. Patients who need hyperglycemic rescue therapy will continue in the trial until they complete all study visits.

Patients who permanently discontinue study drug prior to the Safety Follow-Up Period will also be required to continue in the trial and should receive a new glucose-lowering intervention (Section 7.7.1.2).

Every attempt will be made to keep patients in the trial irrespective of their adherence to treatment with study drug. An ET visit will be conducted for patients who are to be discontinued from the study early (after completing Visit 3 [randomization] and before completing the entire 52-week treatment period). If the patient is discontinued during an unscheduled visit, that visit should be performed as the ET visit. If the patient is discontinued during a scheduled visit, that visit should be performed as the ET visit. In either case, ET visit procedures will be performed according to the Schedule of Activities (Section 2). Patients should be instructed to return any remaining used or unused study drug supplies and patient diaries to the study site at this visit. Study treatment will be terminated at this visit (if not already terminated). Patients will be asked to perform the Safety Follow-Up visit (Visit 801) approximately 30 days after the ET visit, so that the Safety Follow-Up visit will be their final visit. If the patient is not able to perform the Safety Follow-Up visit, then the ET visit will be the final study visit.

Study governance considerations are described in detail in Appendix 3.

5.2. Number of Participants

Approximately 3000 participants will be screened to achieve approximately 1800 randomized patients and approximately 1530 patients completing Week 36 (for an estimated total of 510 patients completing Week 36 per treatment group).

5.3. End of Study Definition

End of the trial is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

Study GBGL utilizes the parallel design as an appropriate option to collect data attributable to a specific randomized therapy without potential interference from other randomized treatments or the randomization sequence. The study is designed to determine the comparative benefits and risks of once weekly dulaglutide 3.0 mg or 4.5 mg versus once weekly dulaglutide 1.5 mg in patients with T2D who cannot maintain glycemic control with metformin alone. To minimize the potential confounding effect of changes to concomitant metformin, patients will be expected to maintain metformin throughout the treatment period until the last dose of randomized treatment, other than for the situations described in Section 7.7.1.1.

The planned duration of treatment for the primary endpoint at 36 weeks is considered appropriate to assess the full effects and benefit/risk of higher doses of dulaglutide compared with the dulaglutide 1.5 mg dose on both glycemic control and body weight. While the full effect of randomized treatments on PG is expected after approximately 8-12 weeks of therapy and on HbA1c after approximately 12-24 weeks (Jendle et al. 2016), a minimum of 26-30 weeks are required to assure maximal effects of dulaglutide on the key secondary outcome of body weight (Davies et al. 2015; Ahrén et al. 2017). The full treatment duration of 52 weeks will allow for collection of additional exposure data to monitor efficacy and safety.

The 2 investigational dulaglutide doses (once weekly 3.0 mg and 4.5 mg) to be studied in this trial were chosen based on simulations which included data collected in Studies H9X-MC-GBCD (GBCD) and GBCF (Barrington et al. 2011; Skrivanek et al. 2014) and evidence for incremental increases in efficacy for both glycemic control and body weight reduction, with an acceptable safety and tolerability profile, from Study GBGJ. Further details and the rationale for the choice of the doses and dose escalation regimens are provided in Section 5.5.

Dulaglutide 1.5 mg is included as the active comparator because it is the maximum approved dose and will enable comparison of efficacy and safety data of higher investigational doses collected under similar conditions. A placebo comparator was not included because it was not considered necessary for addressing the primary or secondary objectives of the trial. Moreover, maintaining patients with inadequate glycemic control on metformin monotherapy, particularly those with relatively poor glycemic control at entry (HbA1c ≥9.0% [75 mmol/mol]), would not be consistent with current recommendations provided in a position statement of the American Diabetes Association and the European Association for the Study of Diabetes (Inzucchi et al. 2015).

5.5. Justification for Dose and Dose Escalation Regimen

The investigational dulaglutide doses (3.0 mg and 4.5 mg) were selected considering adequate separation in PK exposure range between the doses and 1.5 mg to enable evaluation of their efficacy and adverse effect profiles. Ratios of mean area under concentration-time curve (AUC) for 3.0 mg/4.5 mg doses to 1.5 mg dose are 2.26 and 3.10, respectively. The steady-state mean AUC and 90% confidence intervals (CIs) for the 1.5 mg, 3.0 mg, and 4.5 mg doses are 11,800 (5300, 21,300), 26,700 (15,300, 41,400), and 36,600 (21,100, 56,500) ng.h/mL, respectively. Therefore, the ratio of mean AUC for 4.5 mg to 3.0 mg is anticipated to be 1.37. Based on predicted plasma exposures in patients with T2D receiving the highest investigational dulaglutide dose of 4.5 mg, the margins of safety (ie, plasma exposure multiples at the no-observed-adverse-effect level [NOAEL]) in chronic toxicology studies in monkeys and rats were 162-fold and 55-fold, respectively.

Data from Phase 2 Study GBGJ provided evidence for incremental efficacy of dulaglutide 3.0 mg and 4.5 mg once weekly versus 1.5 mg once weekly on glycemic control and body weight reduction. After 18 weeks of treatment (analysis of on-treatment data without rescue), the estimated treatment differences in HbA1c reduction with 3.0 mg and 4.5 mg once weekly relative to 1.5 mg once weekly were -0.22% and -0.26%, respectively. Larger differences were observed in subgroups of patients with higher baseline HbA1c (ie, ≥8.5% [69 mmol/mol] vs. <8.5% [69 mmol/mol]). Dulaglutide 3.0 mg and 4.5 mg once weekly also provided for additional body weight reduction over the 1.5 mg dose that exceeded 1 kg at Week 18, with further weight reduction expected with longer durations of treatment. Although the effects of dulaglutide 4.5 mg were generally greater than the effects of the 3.0 mg dose on measures of glycemic control and body weight, a larger study is required to fully characterize the benefit/risk profile of each higher dulaglutide dose compared with the 1.5 mg dose over a longer duration of treatment.

The dose escalation regimens used for the investigational doses were designed based on evaluation of the algorithms tested in Phase 2 Study GBGJ. While overall tolerability as measured by study drug discontinuations due to AE was comparable between the algorithms, initiating dose escalation at 0.75 mg once weekly generally resulted in a lower overall incidence of GI events. Pharmacokinetic/PD exposure-response modeling of Study GBGJ data also predicted that the daily incidence of nausea and vomiting at higher dulaglutide doses would be lowest in algorithms starting at the lower 0.75 mg dose with a slower dose escalation over an 8-week period, allowing adequate time for tolerance to develop. Based on these findings, the starting dose for all patients in the current study will be 0.75 mg once weekly followed by 1.5 mg once weekly, with a 4-week interval for each dose. Patients will then either remain on 1.5 mg once weekly or escalate to their final assigned dose as shown in Figure GBGL.1 and Table GBGL.2.

6. Study Population

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted.

6.1. Inclusion Criteria

Patients are eligible to be included in the study only if they meet all of the following criteria at study entry (Visit 1) (or at other visits where indicated).

Type of Patient and Disease Characteristics

[1] Patients with T2D for ≥6 months according to the World Health Organization (WHO) classification (Appendix 5) or other locally applicable diagnostic standards, treated with stable doses of metformin (as defined in [4] below) for at least 3 months.

Patient Characteristics

[2] men or nonpregnant women aged ≥18 years;

[3] have HbA1c ≥7.5% (58 mmol/mol) and ≤11.0% (97 mmol/mol), inclusive, as assessed by the central laboratory;

[4] treated with stable doses of metformin for at least 3 months prior to Visit 1 and between Visit 1 and Visit 3; the metformin dose will be considered stable for this period if all prescribed daily doses were in the range between the minimum required dose (≥1500 mg/day) and the maximum approved dose per country-specific label; lower doses will be allowed only with documented GI intolerability in the required dose range or a documented estimated glomerular filtration rate (eGFR; measured by the Chronic Kidney Disease-Epidemiology Collaboration equation [CKD-EPI]) or other renal function measure which requires lower doses per country-specific labeling;

[5] stable body weight for at least 3 months prior to Visit 1 (not changed by more than 5% in the past 3 months);

[6] have a body mass index (BMI) ≥25 kg/m^2;

[7] in the investigator's opinion, are well-motivated, capable, and willing to:

[a] self-inject treatment as required for this protocol or come into the clinic for injections (visually impaired persons who are not able to perform the injections must have the assistance of a sighted individual trained to inject the study drug; persons with physical limitations who are not able to perform the injections must have the assistance of an individual trained to inject the study drug);

[b] perform finger stick PG monitoring at least once daily every day and up to 6 times one day per week at designated times throughout the trial (see Schedule of Activities [Section 2]);

[c] maintain a study diary as required for this protocol;

[8] women of childbearing potential participating must agree to remain abstinent (complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship without sexual relationships with males (as part of their preferred and usual lifestyle), use 1 highly effective method of contraception, or a combination of 2 effective methods of contraception (see Appendix 6) starting at screening and continuing until 4 weeks after the last dose of the randomized therapy. Periodic abstinence (eg, calendar, ovulation, symptothermal, postovulation methods), declaration of abstinence just for the duration of a trial, and withdrawal are not acceptable methods of contraception.

[a] women of childbearing potential participating must test negative for pregnancy as indicated by negative serum pregnancy test at screening (Visit 1) followed by a negative urine pregnancy test within 24 hours prior to exposure (Visit 3).

[b] the patient may choose to use a double-barrier method of contraception. Barrier protection methods without concomitant use of a spermicide are not a reliable or acceptable method. Thus, each barrier method must include use of a spermicide (ie, condom with spermicide, diaphragm with spermicide, female condom with spermicide). It should be noted that the use of male and female condoms as a double-barrier method is not considered acceptable due to the high failure rate when these methods are combined.

[c] women must not be breastfeeding.

[9] women not of childbearing potential may participate and include those who are:

[a] infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as Müllerian agenesis; or

[b] postmenopausal – defined as either

i. A woman ≥50 years of age with an intact uterus, not on hormone therapy who has had either

a) cessation of menses for at least 1 year, or

b) at least 6 months (or longer if required by local regulatory requirements) of spontaneous amenorrhea with a follicle-stimulating hormone >40 mIU/mL (central laboratory, see Schedule of Activities [Section 2]); or

ii. A woman 55 or older not on hormone therapy, who has had at least 6 months of spontaneous amenorrhea; or

iii. A woman at least 55 years of age with a diagnosis of menopause prior to starting hormone replacement therapy.

[10] no male contraception is required except in compliance with specific local government study requirements;

Informed Consent

[11] have given written consent to participate in this study in accordance with local regulations and the Ethics Review Board (ERB) governing the study site

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at study entry (Visit 1), or at other visits, only when indicated:

Medical Conditions

[12] have type 1 diabetes (T1D);

[13] have a history of ≥1 episode of ketoacidosis or hyperosmolar state/coma;

[14] have had ≥1 episode of severe hypoglycemia and/or ≥1 episode of hypoglycemia unawareness within the last 6 months;

[15] have had any of the following CV conditions within 2 months prior to Visit 1: acute myocardial infarction (MI), New York Heart Association Class III or Class IV heart failure, or cerebrovascular accident (stroke);

[16] have a known clinically significant gastric emptying abnormality (eg, severe diabetic gastroparesis or gastric outlet obstruction) or have undergone gastric bypass (bariatric) surgery or restrictive bariatric surgery (eg, Lap-Band®) or chronically take drugs that directly reduce gastrointestinal motility;

[17] have acute or chronic hepatitis, signs and symptoms of any other liver disease other than nonalcoholic fatty liver disease (NAFLD), or alanine aminotransferase (ALT) level >2.5 times the upper limit of the reference range, as determined by the central laboratory at study entry; patients with NAFLD are eligible for participation in this trial;

[18] have had chronic or acute pancreatitis any time prior to study entry;

[19] have an eGFR <30 mL/min/1.73 m^2 (or lower than the country-specific threshold for discontinuing metformin therapy per local label), calculated by CKD-EPI, as determined by the central laboratory at Visit 1 and confirmed at Visit 2;

[20] have a personal or family history of medullary thyroid carcinoma (MTC) or personal history of multiple endocrine neoplasia syndrome type 2;

[21] have serum calcitonin ≥20 ng/L, as determined by the central laboratory at study entry;

[22] have evidence of significant, active autoimmune abnormality (eg, lupus, rheumatoid arthritis) that, in the opinion of the investigator, is likely to require concurrent treatment with systemic glucocorticoids in the next 12 months;

[23] have active or untreated malignancy, or have been in remission from clinically significant malignancy (other than basal cell or squamous cell skin cancer) for less than 5 years;

[24] any serious disease or other condition (eg, known drug or alcohol abuse) which, in the opinion of the investigator, would pose a significant risk to the patient or interfere with the interpretation of safety, efficacy, or PD data;

[25] have any hematologic condition that may interfere with HbA1c measurement (eg, hemolytic anemias, sickle-cell disease);

[26] known proliferative retinopathy or maculopathy requiring acute treatment according to the opinion of the investigator;

Prior/Concomitant Therapy

[27] have used any GLP-1 RAs (including investigational agents with GLP-1 RA activity) or insulin at any time in the past, except for short-term (≤14 consecutive days) use of insulin for acute conditions;

[28] have used any other glucose-lowering medications other than metformin (including insulin) 3 months prior to study entry or during the Screening/Lead-In Periods;

[29] have been treated with any other excluded medication (Section 7.7) within 3 months prior to screening (Visit 1) and/or between study entry and randomization (Visit 3); excluded glucocorticoids must not have been used for >14 days within 1 month prior to Visit 1 or between Visits 1 and 3;

[30] have a condition that is a contraindication for use of the GLP-1 RA class or metformin (per country-specific labels) at Visit 1 or develop such condition between Visit 1 and Visit 3;

[31] discontinuation of metformin therapy, or change in metformin dose or formulation, between Visit 1 and Visit 3;

[32] have been treated with prescription or OTC drugs that promote weight loss (Table GBGL.3) within 3 months prior to screening (Visit 1) and/or between study entry and randomization (Visit 3); or current (or within the last 3 months) participation in, or planned intent to initiate within timeframe of the study, an organized diet and/or exercise weight reduction program other than the lifestyle and dietary measures for diabetes treatment;

Prior/Concurrent Clinical Trial Experience

[33] are currently enrolled in any other clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study;

[34] have participated, within the last 30 days in a clinical trial involving an investigational product. If the previous investigational product has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed;

[35] have previously completed or withdrawn from this study or any other study investigating dulaglutide;

Other Exclusions

[36] are investigator site personnel directly affiliated with this study and/or their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted;

[37] are Lilly employees;

[38] have any other conditions (including known drug or alcohol abuse or psychiatric disorder) or circumstances (including known or anticipated plans for relocation or extended travel) that may preclude the patient from following the protocol and completing the study.

6.3. Lifestyle Restrictions

Per the Schedule of Activities (Section 2), qualified medical staff will provide diabetes management counseling, which will include instructions on diet and exercise and education about the signs, symptoms, and treatment of hypoglycemia, should it occur.

Patients should continue their usual exercise habits and generally follow a healthy meal plan (with consistent meal size and time of day) throughout the course of the study. Dietary counseling may be reviewed throughout the study, as needed. Per Exclusion Criterion [32] (Section 6.2), patients should not initiate during the study an organized diet and/or exercise weight reduction program other than the lifestyle and dietary measures for diabetes treatment.

Study participants should be instructed not to donate blood or blood products during the study.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) must not be rescreened.